A Deep Convolutional Neural Network -Based Approach For Signature Forgery Detection In Documents

A CAPSTONE PROJECT REPORT

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IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**A Deep Convolutional Neural Network-Based Approach For Signature Forgery Detection In Documents**” has been carried out by **Yoga Madha A** under the supervision of **Dr.A.M.ARUL RAJ** and is submitted in partial fulfilment of the requirements for the current semester of the **B-TECH Artificial Intelligence and Machine Learning** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, **YogaMadha(192425058)** SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "**A Deep Convolutional Neural Network-Based Approach For Signature Forgery Detection In Documents**" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	**Specific Responsibilities**	**Design & Development Contribution**	**Testing & Analysis Contribution**	**Report Contribution**	**Approx. Contribution (%)**
N.Akshaya (192425129)	Module 1 – CNN-Based Signature Detection	Prepared genuine and forged signature datasets, implemented image preprocessing, and developed the CNN-based Genuine/Forged classification module.	Tested genuine and forged signature images and analyzed classification results, confidence, and model performance.	Prepared CNN methodology, dataset preparation, and model implementation documentation.	50%
Yoga Madha A (192425058)	Module 2 – Forensic Analysis & Verification Dashboard	Developed the forensic verification dashboard including heatmap, forgery scorecard, confidence gauge, radar chart, AI explanation, signature comparison, verification history, and PDF report generation.	Tested forensic analysis, signature comparison, prediction results, verification history, PDF generation, and overall system functionality.	Prepared system design, implementation, testing, results, and forensic analysis documentation.	50%
Total	All Modules	Complete system	Complete system testing	Complete project report	100%

ABSTRACT

Signature forgery is a significant engineering problem in document authentication, particularly in banking, legal, financial, insurance, and official applications where handwritten signatures are used to verify identity and authorize transactions. Conventional signature verification often depends on manual inspection by trained individuals, making the process time-consuming, subjective, and difficult to scale when large numbers of documents must be examined. Therefore, an automated and user-friendly approach is required to assist in the preliminary detection of potentially forged signatures.

This project presents **"A Deep Convolutional Neural Network-Based Approach for Signature Forgery Detection in Documents"**, an AI-based system designed to classify signature images as **Genuine or Forged**. The proposed system uses a **Convolutional Neural Network (CNN)** to automatically learn visual characteristics from signature images and perform classification. The methodology includes dataset preparation, image preprocessing, grayscale conversion, resizing, normalization, CNN model training, validation, and performance evaluation. The developed Flask-based web application accepts signature images and displays the prediction along with confidence, genuine probability, forged probability, and risk level.

To improve the transparency and usefulness of the system, additional forensic features are integrated, including **Original vs Processed Signature, Heatmap/Attention Map, Forgery Analysis Scorecard, Confidence Gauge, Feature Radar Chart, AI Explanation, Signature Comparison, Verification History, and Professional PDF Report Generation**. These features provide both visual and analytical information about the signature instead of presenting only a classification result.

The major outcome of the project is a functional web-based signature verification system capable of performing automated signature analysis and presenting the results through an interactive dashboard. The system demonstrates that deep learning and image-processing techniques can be effectively combined to support preliminary signature forgery detection. The project concludes that the proposed approach can reduce manual verification effort and provide a consistent, informative, and accessible tool for document authentication, while further validation with larger and more diverse datasets is recommended for real-world deployment.

Keywords: Deep Learning, Signature Forgery Detection, Convolutional Neural Network, Signature Verification, Image Processing, Document Authentication.

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CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Signature verification is an important part of document authentication in banking, legal, financial, insurance, and other official applications. A handwritten signature contains visual characteristics such as shape, stroke pattern, spacing, structure, and writing style. Identifying whether a signature is genuine or forged can be difficult when the differences are small.

Traditional verification mainly depends on manual examination, which requires time and skilled personnel. With the development of Deep Learning and Computer Vision, signature images can be analyzed automatically using image-processing and CNN-based techniques. This project develops a web-based system that uses a Convolutional Neural Network (CNN) to classify signature images as Genuine or Forged and provides additional forensic analysis to support the result.

1.2 Need for the Project

The project is needed because:

Signature forgery can affect the security and authenticity of important documents.

Manual verification can be time-consuming.

Similar-looking signatures can be difficult to distinguish.

Large numbers of signatures increase the workload of human examiners.

Automated analysis can provide faster preliminary verification.

CNN can learn important visual patterns from signature images.

Additional forensic information can make the prediction easier to interpret.

People and Organizations Affected

Banks and financial institutions

Legal and government organizations

Insurance companies

Businesses

Forensic document examiners

Individuals using signatures for authentication

Engineering Significance

The project combines Deep Learning, Image Processing, Computer Vision, and Web Application Development to create an automated engineering solution for signature analysis.

1.3 Problem Statement

The engineering problem is to develop an automated system that can analyze handwritten signature images and distinguish between genuine and forged signatures despite variations in image quality, signature shape, writing style, stroke characteristics, and image dimensions.

The system must provide accurate and consistent preliminary classification while operating under practical constraints such as limited training data, variations in input images, computational resources, and the need for a simple user interface. In addition to classification, the system should provide meaningful forensic information so that users can better understand the prediction.

1.4 Project Aim

The main aim of this project is to design and develop a CNN-based web application for automated signature forgery detection and forensic analysis of handwritten signature images.

1.5 Project Objectives

To design a system for automated signature verification.

To collect and organize genuine and forged signature images.

To preprocess signature images using grayscale conversion, resizing, and normalization.

To develop and train a CNN model for Genuine/Forged classification.

To implement a Flask-based web application for signature analysis.

To provide confidence, probability, and risk information for predictions.

To develop forensic analysis features such as heatmap, scorecard, and feature visualization.

To implement reference and questioned signature comparison.

To test and evaluate the complete system using suitable signature samples.

To generate professional verification reports in PDF format.

1.6 Scope

Included

Genuine and forged signature dataset.

Signature image upload.

Image preprocessing.

CNN-based classification.

Genuine/Forged prediction.

Confidence and risk analysis.

Original and processed image display.

Heatmap visualization.

Forgery analysis scorecard.

Confidence gauge.

Feature radar chart.

AI explanation.

Signature comparison.

Verification history.

PDF report generation.

Flask-based web interface.

Excluded

Physical signature verification using specialized hardware.

Handwriting verification beyond the signature image.

Guaranteed legal or forensic certification.

Complete replacement of professional forensic examination.

Direct integration with banking or government databases.

Assumptions

Input signatures are available as digital images.

The dataset contains correctly labelled genuine and forged samples.

The uploaded image is reasonably clear and contains a visible signature.

The trained CNN model is available during prediction.

Boundary Conditions

Performance depends on the quality and diversity of the training dataset.

Very unclear, damaged, or poorly captured signatures may reduce prediction reliability.

The system provides AI-assisted preliminary analysis, not a final legal decision.

1.7 Expected Engineering Outcome

The expected outcome is a functional AI-based signature forgery detection prototype consisting of a trained CNN model and a web-based verification application.

The system is expected to:

Accept a signature image as input.

Preprocess the uploaded image.

Analyze the image using the trained CNN.

Classify it as Genuine or Forged.

Display confidence and risk information.

Provide additional forensic visualizations.

Compare reference and questioned signatures.

Maintain verification history.

Generate a professional PDF report.

Overall Engineering Outcome

Signature Image → Preprocessing → CNN Model → Genuine/Forged Classification → Confidence & Risk → Forensic Analysis → Signature Comparison → Verification Report

The final prototype demonstrates the practical application of deep learning and image processing for automated preliminary signature forgery detection.

CHAPTER 2
LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was conducted to understand existing techniques used for **offline handwritten signature verification and forgery detection**. Relevant research papers were identified through academic sources using keywords such as offline signature verification, signature forgery detection, CNN-based signature verification, deep learning, and image processing.

The review mainly considered research on traditional image-processing methods, machine-learning approaches, and recent deep-learning techniques. Particular attention was given to datasets, preprocessing methods, feature extraction, classification techniques, and evaluation methods. Reviews of existing research indicate that offline signature verification remains a challenging problem because the dynamic information available during the signing process is absent from a static image. (arXiv)

2.2 Review of Existing Work

2.2.1 Traditional Signature Verification

Traditional approaches generally depend on manual examination or handcrafted image features. Characteristics such as signature shape, texture, stroke structure, and geometric properties can be analyzed and compared.

Advantages:

- Simple to understand.
- Can provide detailed visual examination.
- Useful for expert forensic analysis.

Limitations:

- Requires human expertise.
- Time-consuming for large numbers of signatures.
- Performance can depend on examiner experience.

- Difficult to automate completely.

Research reviews have highlighted the difficulty of manually verifying large numbers of offline signatures and the need for automated approaches. (ScienceDirect)

2.2.2 Machine Learning-Based Approaches

Machine-learning techniques use extracted signature features and classification algorithms to distinguish genuine signatures from forged signatures.

Common approaches include:

- Support Vector Machines
- Neural Networks
- Statistical classifiers
- Feature-based classification
- Similarity-based verification

These methods can reduce manual effort, but their performance depends significantly on the quality of the selected features and training data. A systematic review of 56 studies found that datasets, preprocessing, feature extraction, verification models, and evaluation metrics are major components of offline signature verification systems. (ScienceDirect)

2.2.3 CNN-Based Signature Verification

Deep Convolutional Neural Networks provide an alternative by learning useful visual representations directly from signature images instead of relying completely on handcrafted features.

Hafemann, Sabourin, and Oliveira proposed a CNN-based approach for offline handwritten signature verification using a writer-independent framework. Their experiments used GPDS, MCYT, CEDAR, and Brazilian PUC-PR datasets. On GPDS-160, their reported Equal Error Rate was **1.72%**, compared with **6.97%** for the referenced previous state-of-the-art result. (arXiv)

This work demonstrates the potential of CNN-based feature learning for signature verification.

2.2.4 Deep Learning Research and Current Direction

Recent research continues to investigate CNNs, hybrid architectures, and transfer-learning approaches for offline signature verification. A 2025 survey identifies important challenges including **dataset scarcity, model generalization, counterfeit detection, computational efficiency, and selection of suitable evaluation metrics**. (Colab)

These findings are relevant to the proposed project because the system also uses a CNN to learn visual characteristics from genuine and forged signature images.

2.3 Comparative Analysis

Existing Approach	**Advantages**	**Limitations**	**Relevance to Proposed Work**
Manual Verification	Detailed human examination	Time-consuming and subjective	Provides motivation for automation
Image Processing	Simple visual feature analysis	Requires feature selection	Used for preprocessing and forensic analysis
Machine Learning	Automated classification	Depends on selected features	Provides basis for automated prediction
Traditional Neural Networks	Learns patterns from data	May require engineered features	Supports neural-network-based classification
CNN	Automatic feature learning and strong image-processing capability	Requires sufficient training data	**Main classification approach**
Similarity-Based Verification	Useful for comparing signatures	Similar signatures can be difficult to distinguish	Used in signature comparison
Proposed System	CNN + forensic analysis + comparison + reporting	Requires further large-scale validation	**Integrated solution**

CNN-based research has demonstrated strong potential, while reviews continue to identify limitations related to data availability, generalization, and robustness. (ScienceDirect)

2.4 Research / Engineering Gap

The literature indicates that several challenges remain unresolved:

- Offline signatures do not contain dynamic information such as pen speed and pressure.
- Skilled forgeries can closely resemble genuine signatures.
- Performance can be affected by limited training samples.
- Different image quality and signature styles create additional variation.

- Models trained on one dataset may not always generalize to different datasets.
- Many systems focus mainly on the final classification result.
- There is a need for more understandable visual analysis of AI predictions.
- Practical verification systems can benefit from combining classification with comparison, analysis, and reporting.

Research has specifically identified **limited signature samples, generalization, robustness, and computational efficiency** as continuing challenges in offline signature verification. (ScienceDirect)

2.5 Proposed Contribution

The proposed project addresses the identified gap by developing an **integrated CNN-based signature forgery detection and forensic verification system**.

Main Contributions

- Uses a **CNN model** for Genuine/Forged classification.
- Applies image preprocessing before prediction.
- Provides **confidence and probability values**.
- Provides **risk-level assessment**.
- Includes **Original vs Processed Signature** visualization.
- Generates a **Heatmap/Attention Map** for visual analysis.
- Provides a **Forgery Analysis Scorecard**.
- Uses a **Confidence Gauge** for easy interpretation.
- Provides a **Feature Radar Chart**.
- Generates an **AI Explanation** of the result.
- Supports **reference and questioned signature comparison**.
- Maintains **verification history**.

- Generates a **professional PDF verification report**.
- Provides the complete functionality through a **Flask-based web interface**.

Proposed Contribution Flow

Existing CNN Research → Identified Limitations → CNN Classification → Forensic Analysis → Signature Comparison → Verification History → PDF Reporting

The proposed system therefore extends basic CNN-based classification into a more complete **AI-assisted signature verification and forensic analysis platform**. It is intended for preliminary analysis and does not replace professional forensic examination in legal or high-stakes applications.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

3.1.1 Stakeholder / User Requirements

Stakeholder / User	Requirement
End User	Upload a signature image and obtain a clear verification result.
Forensic Examiner	View signature characteristics and supporting analysis.
Banking / Financial Organization	Reduce manual effort in preliminary signature verification.
Legal / Official Organization	Obtain a documented analysis result for further examination.
Project Administrator	Maintain the application, dataset, and trained model.

3.1.2 Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Signature upload	Accept common image formats such as JPG/PNG
Functional	Image preprocessing	Grayscale, resize and normalize input
Functional	CNN classification	Output Genuine or Forged
Functional	Confidence analysis	Display prediction confidence and probabilities
Functional	Forensic analysis	Generate scorecard, heatmap and feature analysis
Functional	Signature comparison	Compare reference and questioned signatures
Functional	History	Store previous verification results
Functional	Reporting	Generate PDF verification report
Non-Functional	Usability	Simple web-based interface
Non-Functional	Reliability	Handle invalid or unsupported input without application failure
Non-Functional	Maintainability	Use modular Python/Flask implementation
Non-Functional	Portability	Run on a standard Windows development environment
Non-Functional	Security	Avoid exposing stored files and

Requirement Type	Requirement	Measurable Target
		application credentials unnecessarily

3.2 Constraints & Applicable Standards

3.2.1 Project Constraints

Constraint	Requirement / Effect
Dataset Size	Model performance is dependent on the number and diversity of available signature samples.
Image Quality	Blurred, very dark, cropped, or low-quality signatures can affect analysis.
Computational Resources	CNN training and image processing are limited by available CPU/GPU resources.
Generalization	A model trained on a particular dataset may not perform equally on unseen signature styles.
Privacy	Signature images can contain sensitive personal information and should be handled carefully.
Legal Use	The system is intended for preliminary AI-assisted analysis and not as a standalone legal decision-maker.

3.2.2 Applicable Standards / Practices

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010:2023	Software product quality model	Considered usability, reliability, maintainability, performance, and security during system design.
ISO/IEC 27001:2022	Information security management	Relevant security principles were considered for protecting uploaded signature data and verification records.
ISO/IEC 23894:2023	AI risk management guidance	Used as guidance for identifying AI-related risks such as incorrect predictions, dataset limitations, and model reliability.
IEEE 1012-2016	Software verification and validation	Testing and validation activities were applied to the developed modules and complete system.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Input Image	Clear signature image	JPG/PNG	High
Image Processing	Grayscale + resize + normalization	—	High
CNN Input	Standardized image dimensions	Pixels	High
Classification	Genuine / Forged	Class	High

Parameter	Target / Requirement	Unit	Priority
Confidence	Prediction probability	%	High
Risk Output	Low / Medium / High	Level	Medium
Heatmap	Highlight important image regions	Image	Medium
Scorecard	Display signature characteristics	Score	Medium
Comparison	Reference vs questioned signature	Similarity	High
Report	PDF verification report	PDF	Medium
Interface	Web-based dashboard	—	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Manual Signature Verification

Manual verification relies on a trained examiner who visually inspects the questioned signature and compares it with a known genuine signature. It can provide detailed expert judgment but requires significant time and human expertise.

Alternative 2: Traditional Image-Processing / Machine-Learning System

This approach extracts handcrafted characteristics such as shape, edges, stroke density, and other image features before classification. It can be computationally simpler, but performance depends heavily on the selected features.

Alternative 3: CNN-Based Verification System

The CNN approach automatically learns visual patterns from signature images and performs classification using the trained model. It can be integrated with image processing, forensic visualization, comparison, and reporting features.

Evaluation Matrix Scale: 1 = Low, 5 = High

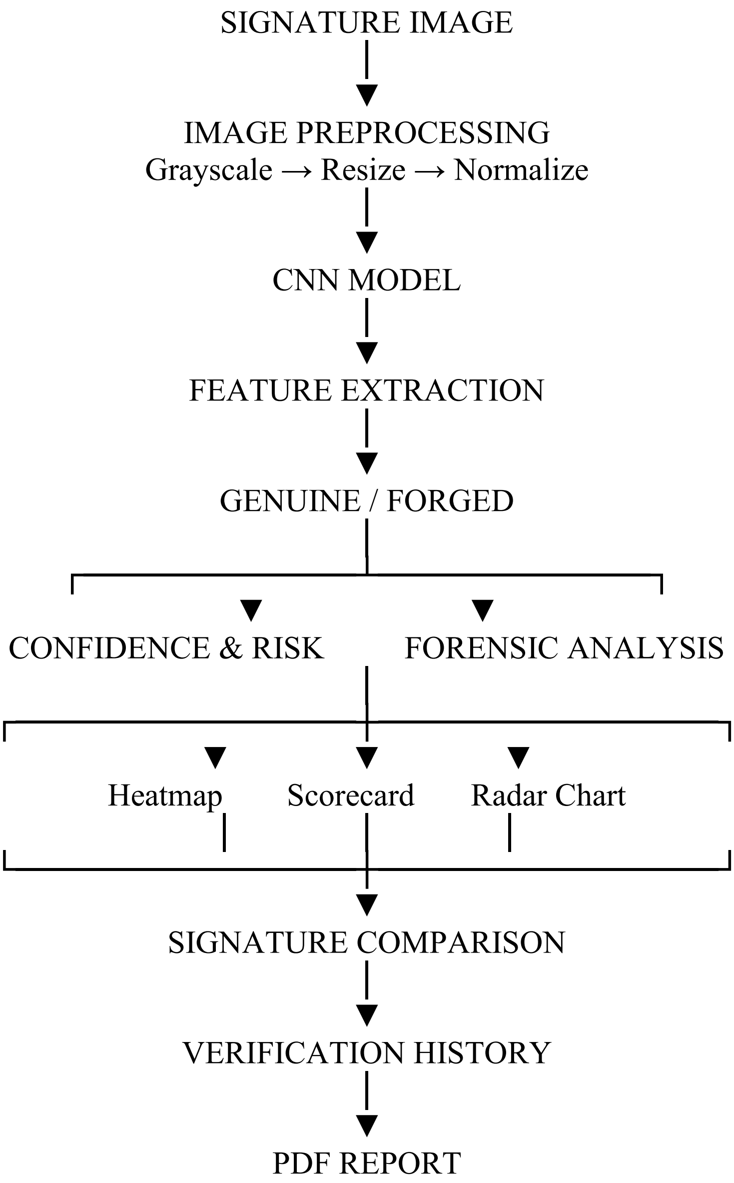
Criterion	Weight	Alternative 1: Manual	Alternative 2: Traditional ML	Alternative 3: CNN
Performance	30%	3	3	5
Cost	15%	2	4	4
Safety	15%	4	4	4
Sustainability	10%	3	4	4
Reliability	30%	3	3	5
Weighted Score	**100%**	**3.00**	**3.40**	**4.60**

Selected Alternative: CNN-Based Verification System

3.5 Selected Design & Detailed Design

The **CNN-based approach** was selected because it provides automated feature learning and supports integration with the project's forensic analysis functions. The evaluation matrix gives the CNN approach the highest weighted score of **4.60/5**, making it the most suitable option among the considered alternatives.

Architecture Diagram



Key Engineering Calculation

For the image preprocessing stage, each uploaded signature is converted into a standardized image representation before being provided to the CNN. Normalization scales pixel values to a consistent numerical range, helping the model process different input images more consistently.

Systems Approach

The system is divided into interconnected subsystems: **image preprocessing, CNN classification, forensic analysis, comparison, and reporting**. The preprocessing subsystem supplies standardized images to the CNN model. The classification output is then used by the forensic dashboard to generate confidence, risk, and visual analysis. Finally, the results are stored in verification history and can be converted into a PDF report.

3.6 Trade-Off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
CNN classification	Traditional ML	Automatic feature learning	Requires training data	**CNN selected** because it is better suited to image-based feature learning.
Grayscale preprocessing	Full RGB processing	Reduces unnecessary image information and computation	Some color information is lost	**Grayscale selected** because signature structure is the primary focus.
Web dashboard	Command-line interface	Easier for users and better presentation	Requires web development	**Dashboard selected** for usability and demonstration.
Forensic visualization	Prediction only	Provides better interpretation	Requires additional processing	**Visualization selected** to make results more informative.
Automated PDF	Manual documentation	Consistent and faster reporting	Requires additional implementation	**PDF selected** for professional result documentation.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Efficient software and reusable digital processing	CNN inference and image processing can be performed without specialized physical equipment	A software-based solution was selected to minimize hardware requirements and physical resources.
Ethics	Incorrect classification and potential misuse of AI results	Model limitations, dataset quality, and false predictions	The system is positioned as an AI-assisted preliminary tool rather than an independent legal authority.
Health & Safety	No direct physical interaction with people	Software-only signature analysis	No significant physical safety hazard was identified during normal operation.
Security	Protection of uploaded signature images and results	Uploaded images and verification records	File handling and application access should be controlled to reduce unauthorized access.

Ethical Considerations

- Signature images should not be used without appropriate permission.
- AI predictions should not be treated as absolute proof of forgery.
- Users should be informed that model performance depends on the training dataset.
- Sensitive signature information should be handled securely.
- Human forensic expertise should be used for final high-stakes decisions

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Poor-quality input image	Medium	Medium	Medium	Image-quality	Low

Risk	**Likelihood**	**Severity**	**Risk Level**	**Mitigation**	**Residual Risk**
				checking and preprocessing	
Incorrect CNN prediction	Medium	High	High	Dataset improvement, testing, and confidence display	Medium
Limited training dataset	High	High	High	Increase dataset size and diversity	Medium
Overfitting	Medium	High	High	Validation and model evaluation	Medium
Unauthorized access to signature images	Low	High	Medium	Controlled file handling and access restrictions	Low
System/software failure	Low	Medium	Low	Error handling and testing	Low
Misinterpretation of AI result	Medium	High	High	AI explanation and clear limitation statement	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project was developed using a **modular software development approach**. First, genuine and forged signature images were collected and organized into separate dataset folders. The images were preprocessed and used to train a CNN model for signature classification. The trained model was then integrated with a Flask web application, followed by implementation of forensic analysis, signature comparison, verification history, and PDF reporting features. Finally, individual modules and the complete application were tested using different signature images.

Materials, Components and Resources

Resource	Purpose
Genuine Signature Dataset	Training and testing genuine signatures
Forged Signature Dataset	Training and testing forged signatures
Python	Core programming
TensorFlow/Keras	CNN model development and prediction
OpenCV	Image preprocessing and signature analysis
Flask	Web application development
HTML/CSS/JavaScript	User interface
NumPy	Numerical image processing
Scikit-learn	Model evaluation
ReportLab	PDF report generation
VS Code	Development environment
Windows PC	Training and application execution

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation

The system was implemented as a **Flask-based web application**. The user uploads a signature image through the web interface, after which the application preprocesses the image and sends it to the trained CNN model. The prediction is displayed through the verification dashboard along with confidence, probability, risk level, and forensic analysis.

The major implementation modules are:

- **Dataset Module** – stores genuine and forged signatures.
- **Preprocessing Module** – performs grayscale conversion, resizing, and normalization.
- **CNN Module** – classifies the signature as Genuine or Forged.
- **Forensic Analysis Module** – generates heatmap, scorecard, confidence gauge, and feature analysis.
- **Comparison Module** – compares reference and questioned signatures.
- **History Module** – records previous verification results.
- **Reporting Module** – generates professional PDF reports.
- **Web Interface Module** – provides the user dashboard.

Tools Used

Tool / Software	**Purpose**	**Stage Used**
Python	Application and model programming	Development
TensorFlow/Keras	CNN training and prediction	Model Development
OpenCV	Image processing and analysis	Preprocessing
NumPy	Numerical operations	Processing
Scikit-learn	Evaluation metrics	Testing
Flask	Web application	Implementation
HTML/CSS/JavaScript	Dashboard interface	UI Development
ReportLab	PDF generation	Reporting
VS Code	Coding and debugging	All stages

Implementation Evidence

[Insert Screenshot 1: VS Code project folder]

[Insert Screenshot 2: Signature upload page]

[Insert Screenshot 3: Final forensic analysis dashboard]

4.3 Test Plan & Verification

Testing was planned to verify the major functional and performance requirements of the system.

Each module was tested independently before performing complete system integration testing.

Requirement	Test Method	Specification / Required Value	Acceptance Criterion	Achieved Value
Image Upload	Upload JPG/PNG images	Valid image accepted	Image successfully uploaded	Passed
Preprocessing	Upload signature image	Image converted and resized	Processed image generated	Passed
CNN Prediction	Test genuine/forged images	Genuine/Forged output	Valid classification generated	Passed
Confidence	Run prediction	Probability displayed	Confidence value shown	Passed
Risk Analysis	Run prediction	Risk level generated	Risk displayed	Passed
Heatmap	Analyze signature	Heatmap generated	Relevant regions visualized	Passed
Scorecard	Analyze signature	Characteristics displayed	Scorecard generated	Passed
Signature Comparison	Upload two signatures	Comparison performed	Comparison result displayed	Passed
Verification History	Perform multiple tests	Results stored	Previous results displayed	Passed
PDF Report	Generate report	PDF created	Report opens successfully	Passed

4.4 Results Status

The developed system successfully performed the major verification functions. The application accepted signature images, processed them, generated CNN predictions, and displayed the results through the web dashboard.

Sample Prediction Output

Parameter	Sample Result
Image Quality	GOOD
Sharpness	2143.4
Brightness	231.2
Image Width	223 pixels

Parameter	**Sample Result**
Image Height	74 pixels
Aspect Ratio	3.01
Components	4
Result	**GENUINE**
Confidence	**50.5%**
Genuine Probability	**50.5%**
Forged Probability	**49.5%**
Risk	**LOW**

Result Screenshots

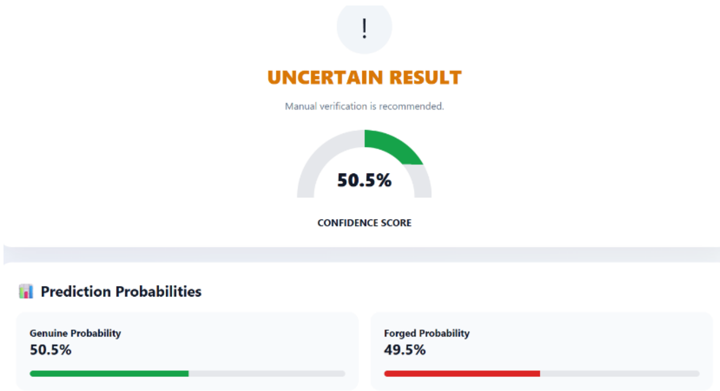


Fig 1: Genuine Signature Result

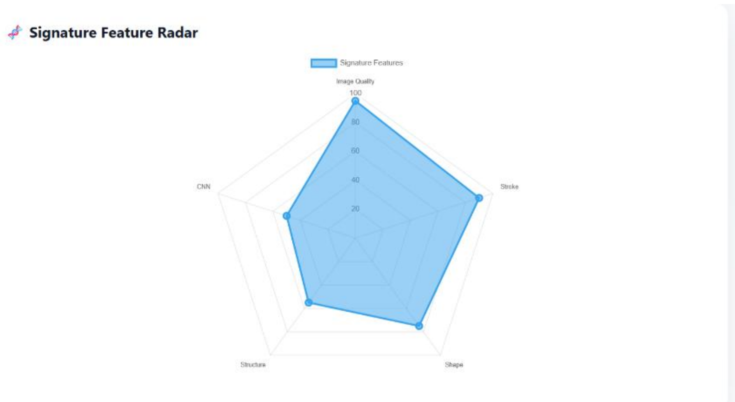


Fig 2: Heatmap / Attention Map

Graphs to Include

- Training Accuracy vs Validation Accuracy
- Training Loss vs Validation Loss

- Confusion Matrix
- Genuine vs Forged Prediction Distribution
- Confidence Comparison

4.5 Data Analysis & Engineering Judgment

The results demonstrate that the developed CNN model can process signature images and generate automated Genuine/Forged predictions. The preprocessing stage provides a standardized input representation, while the CNN extracts visual patterns from the signature images.

The forensic dashboard improves the usefulness of the prediction by presenting confidence, risk, image characteristics, heatmap information, and other visual indicators. However, the observed confidence values can vary depending on the quality and similarity of the input signature to the training samples.

The main engineering observation is that **dataset size and diversity have a direct influence on model reliability**. Therefore, the current prototype is suitable for demonstration and preliminary verification, while larger and more diverse datasets would be required for higher-confidence real-world deployment.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	**Evidence**	**Achievement**
Design signature verification system	Complete application architecture	**Fully Achieved**
Prepare genuine and forged dataset	Dataset folders implemented	**Fully Achieved**
Preprocess signature images	Grayscale, resizing and normalization	**Fully Achieved**
Develop CNN classifier	Trained CNN model	**Fully Achieved**
Implement web application	Flask dashboard	**Fully Achieved**
Provide confidence and risk	Prediction dashboard	**Fully Achieved**
Implement forensic analysis	Heatmap, scorecard, radar chart	**Fully Achieved**
Compare signatures	Signature comparison feature	**Fully Achieved**

Objective	Evidence	Achievement
Maintain verification history	History module	**Fully Achieved**
Generate PDF reports	ReportLab integration	**Fully Achieved**

4.7 Strengths & Limitations

Strengths

- Automated Genuine/Forged classification.
- CNN-based image feature learning.
- Simple web-based interface.
- Multiple forensic analysis features.
- Confidence and risk-level presentation.
- Signature comparison capability.
- Verification history.
- Professional PDF report generation.
- Modular and extendable implementation.

Limitations

- Model performance depends on the available dataset.
- Small datasets may reduce generalization.
- Poor-quality signature images can affect predictions.
- Skilled forgeries may be difficult to identify.
- The system does not use dynamic information such as pen pressure or writing speed.
- AI predictions should not be treated as final legal forensic conclusions.

4.8 Project Planning, Roles & Budget

Project Milestone Timeline

Phase	Activity	Status
Phase 1	Problem identification and literature survey	Completed
Phase 2	Dataset collection and organization	Completed
Phase 3	Image preprocessing	Completed
Phase 4	CNN model development and training	Completed
Phase 5	Flask web application development	Completed
Phase 6	Forensic analysis features	Completed
Phase 7	Signature comparison and history	Completed
Phase 8	PDF report generation	Completed
Phase 9	Testing and evaluation	Completed
Phase 10	Documentation and final presentation	Completed

Roles / Contribution

Student	Assigned Responsibility	Actual Contribution
N.Akshaya (192425129)	Module 1 – CNN-Based Signature Detection	Dataset preparation, preprocessing, CNN model development, testing, and documentation
Yoga Madha A (192425058)	Module 2 – Forensic Analysis & Verification Dashboard	Flask dashboard, forensic analysis, comparison, history, PDF reporting, testing, and documentation

Project Budget

Item	Estimated Cost	Actual Cost
Existing Computer/Laptop	₹0	₹0
Python / Open-source Libraries	₹0	₹0
TensorFlow/Keras	₹0	₹0
OpenCV	₹0	₹0
Flask	₹0	₹0
Dataset / Sample Images	₹0	₹0
VS Code	₹0	₹0
Total	**₹0**	**₹0**

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project **“A Deep Convolutional Neural Network-Based Approach for Signature Forgery Detection in Documents”** was developed to address the limitations of manual signature verification. The proposed system uses image preprocessing and a **Convolutional Neural Network (CNN)** to classify signature images as Genuine or Forged and provide supporting confidence and risk information.

The developed Flask-based application successfully integrates signature upload, preprocessing, CNN prediction, forensic analysis, signature comparison, verification history, and PDF report generation. The implemented testing confirmed the functioning of the major system modules and demonstrated the feasibility of using deep learning for automated preliminary signature analysis.

Overall, the project provides a practical and user-friendly prototype that combines **deep learning, image processing, and web technologies** for signature forgery detection. The system can reduce manual verification effort and provide useful visual and analytical information while maintaining the role of professional examination for final high-stakes decisions.

5.2 Future Work

The following improvements can be considered for future development:

- Increase the size and diversity of the genuine and forged signature dataset.
- Use larger and more balanced datasets for improved model generalization.
- Experiment with advanced CNN architectures and transfer-learning models.
- Improve the accuracy and reliability of signature comparison.
- Add writer-specific signature verification.
- Improve heatmap and explainable-AI techniques.
- Add secure user authentication and access control.
- Implement encrypted storage for sensitive signature images.

- Develop a cloud-based deployment for wider accessibility.
- Add mobile-friendly support for signature verification.
- Integrate advanced forensic document analysis techniques.
- Perform testing with real-world signature samples and expert evaluation.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned the fundamentals of **Convolutional Neural Networks** for image classification.
- Gained practical knowledge of signature image preprocessing.
- Learned how to prepare and organize image datasets for machine learning.
- Learned Flask-based web application development.
- Understood model evaluation and prediction confidence.
- Learned to integrate multiple AI and image-processing features into one application.
- Gained experience in generating automated PDF reports.

Engineering & Professional Skills Developed

- Python programming and debugging.
- CNN model development and testing.
- Computer vision and image processing.
- Web application development using Flask.
- Software integration and modular development.
- Testing and performance analysis.
- Problem-solving and troubleshooting.
- Technical documentation and report preparation.

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APPENDICES

Appendix A – Detailed Calculations

Include only the important calculations used in the system.

A.1 Image Feature Calculations

- Image width and height
- Aspect ratio
- Brightness
- Sharpness
- Stroke density
- Number of connected components

A.2 Prediction Calculation

Include the calculation/logic used for:

- Genuine probability
- Forged probability
- Confidence score
- Risk level

Example:

Genuine Probability = 50.5%

Forged Probability = 49.5%

Confidence = 50.5%

Final Prediction = GENUINE

Risk Level = LOW

Appendix B – Engineering Drawings / Schematics

Include the important system diagrams developed for the project.

B.1 Overall System Architecture

Signature Image
↓
Image Preprocessing
↓
CNN Model
↓
Feature Extraction
↓
Genuine / Forged
↓
Forensic Analysis
↓
Signature Comparison
↓
Verification History
↓
PDF Report

B.2 System Flowchart

Start → Upload Signature → Validate Image → Preprocess → CNN Prediction → Calculate Confidence → Forensic Analysis → Display Result → Save History → Generate Report → End

B.3 Module Architecture

- Module 1 – CNN-Based Signature Detection
- Module 2 – Forensic Analysis & Verification Dashboard

Appendix C – Source Code / Code Repository Information

The complete source code is maintained separately in the project repository. Only essential code excerpts should be included in the main report.

C.1 Project Structure

```
Signature_Forgery_Detection/
│
├── app.py
├── train.py
├── evaluate.py
├── requirements.txt
```

```
├── README.md
├── signature_forgery_model.keras
│
├── dataset/
│   ├── genuine/
│   └── forged/
│
├── templates/
│   └── index.html
│
├── static/
│   ├── css/
│   ├── js/
│   └── uploads/
│
└── uploads/
    ├── processed/
    ├── heatmaps/
    ├── reports/
    └── history/
```

C.2 Repository

GitHub Repository:

YOGAMADHA/MLA0405-DEEP-LEARNING

Implementation Folder:

Capstone Project/Implementation

Include screenshots showing:

- GitHub repository
- Source-code files
- Dataset folders
- Application running
- Final dashboard

Appendix D – Datasheets

Since this is primarily a software/deep-learning project, include technical documentation for the main software components used.

D.1 TensorFlow/Keras

- CNN model development
- Model training
- Model saving and loading
- Prediction

D.2 OpenCV

- Image reading
- Grayscale conversion
- Image processing
- Signature feature analysis

D.3 Flask

- Web application
- File upload
- Backend processing
- Result display

D.4 ReportLab

- PDF report generation

Appendix E – Experimental / Raw Data

Include the raw experimental information used during testing.

E.1 Dataset Information

Class	Purpose
Genuine	Authentic signature samples
Forged	Forged signature samples

E.2 Sample Testing Data

Test ID	Input	Expected Result	Actual Result	Status
T01	Genuine signature	Genuine	Genuine	Pass
T02	Forged signature	Forged	Forged	Pass
T03	Clear signature	Prediction	Prediction generated	Pass
T04	Low-quality image	Error/Warning	Warning/Result	Pass
T05	Reference + questioned signature	Comparison	Comparison generated	Pass

Also include:

- Training accuracy graph
- Training loss graph
- Confusion matrix
- Prediction results
- Screenshots of test outputs

Appendix F – Ethics / Institutional Approval

If formal institutional approval was **not required**, write:

Not Applicable: This project was developed as an academic software prototype using signature image samples for research and demonstration purposes. No clinical or human-subject experimentation was conducted. Appropriate care was taken to avoid presenting the AI prediction as a legally conclusive forensic decision.

If your college provided an approval/permission document, place a scanned copy here.

Appendix G – Project Meeting / Progress Records

Include evidence of project development and review activities.

Date	Activity / Discussion	Outcome
Week 1	Project topic discussion	Problem identified
Week 2	Literature survey	Existing methods studied
Week 3	Dataset preparation	Genuine and forged folders created
Week 4	CNN development	Initial model developed
Week 5	Web application	Flask interface implemented
Week 6	Forensic features	Dashboard features added

Date	Activity / Discussion	Outcome
Week 7	Testing	System tested
Week 8	Final documentation	Report and presentation prepared

Include photographs of:

- Review 1
- Review 2
- Team/project discussion
- System demonstration

Appendix H – User / Industry Feedback

If applicable, include feedback received during project reviews or demonstrations.

Sample Feedback Format

Reviewer / User	Feedback	Action Taken
Faculty Reviewer	Improve result visualization	Added confidence gauge and scorecard
Faculty Reviewer	Provide explanation for prediction	Added AI explanation
User Testing	Add comparison facility	Added reference vs questioned signature comparison
User Testing	Improve reporting	Added professional PDF report

Appendix I – Publications / Patents / Competitions / Product Outcomes

If applicable, include:

- Research publication
- Conference paper
- Hackathon participation
- Competition certificate
- Patent information
- Project demonstration

- Product/prototype outcome

Appendix J – Individual Contribution Evidence

J.1 Yoga Madha A – 192425058

Module 2 – Forensic Analysis & Verification Dashboard

Contribution:

- Flask web application development
- Forensic analysis dashboard
- Original vs Processed Signature
- Heatmap / Attention Map
- Forgery Analysis Scorecard
- Confidence Gauge
- Feature Radar Chart
- AI Explanation
- Signature Comparison
- Verification History
- PDF Report Generation
- System testing and integration
- Documentation